

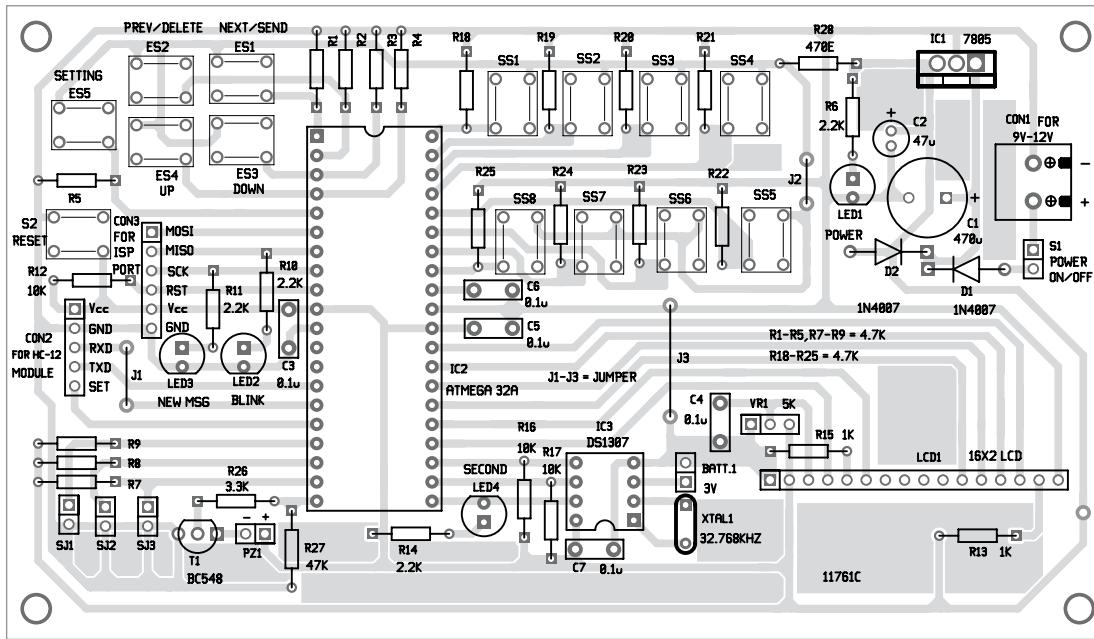


Fig. 3: Components layout of the PCB

PARTS LIST

Semiconductors:

IC1	- 5V voltage regulator, 7805
IC2	- ATmega32A microcontroller
IC3	- DS1307 RTC
LED1-LED4	- 5mm LED
T1	- BC548 npn transistor
D1, D2	- 1N4007 rectifier diode
HC-12 module	- 5-pin wireless transceiver with antenna

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1-R5, R7-R9,	- 4.7-kilo-ohm
R18-R25	- 4.7-kilo-ohm
R6, R10, R11,	- 2.2-kilo-ohm
R14	- 2.2-kilo-ohm
R12, R16, R17	- 10-kilo-ohm
R13, R15	- 1-kilo-ohm
R26	- 3.3-kilo-ohm
R27	- 47-kilo-ohm
R28	- 470-ohm
VR1	- 5-kilo-ohm pot

Capacitors:

C1	- 470 μ F, 25V electrolytic
C2	- 47 μ F, 16V electrolytic
C3-C7	- 0.1 μ F ceramic disk

Miscellaneous:

CON1	- 2-pin terminal connector
CON2	- 5-pin connector
CON3	- 6-pin connector
SJ1-SJ3	- 2-pin connector with shorting jumper
S1	- On/off switch
S2, ES1-ES5,	- Tactile switch
SS1-SS8	- 32.768kHz crystal
XTAL1	- 3V battery
BATT.1	- 16x2 LCD module
LCD1	- Piezo buzzer
PZ1	- 12V DC power supply
	- Jumper wires for J1-J3
	- 20-pin Bergstrip (male)
	- 40-pin IC base

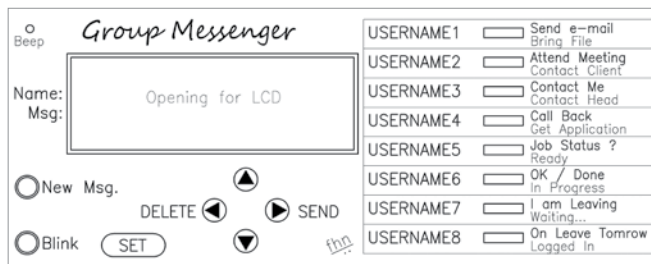


Fig. 4: Proposed panel for group messenger

and loop the number) to set current date and time. If no button is pressed for more than five seconds the edited/current date and time gets saved to the RTC IC DS1307.

At least two such modules (PCBs) are required to send and receive a message in the project. A maximum of eight modules can be used to communicate with each other.

Construction and testing

An actual-size PCB layout for the group messenger is shown in Fig. 2 and its components layout in Fig. 3. Assemble the circuit on the PCB. Connect 9V or 12V DC power supply across CON1 and 3V battery across BATT.1. Before inserting MCU into the PCB, do not forget to upload *GroupMessenger.hex* into it.

Use at least two modules for testing; one (Module1) as transmitter and second (Module2) as receiver. The author's proposed panel for

group messenger is shown in Fig. 4.

Sending a message

Buttons SS1 through SS8 are selection buttons to select username, first set of the message and second set of the message consecutively. The same eight buttons are used for selecting usernames and two messages assigned to each button. Each button is used for one username and two messages, which are selected by pressing the same button consecutively.

To send a message, first select the username (actual user ID number of SET2 or receiver)

by pressing any one of the selection buttons SS1 through SS8. The selected name is displayed on LCD1. Press the same button again to send the required message.

The username and ID are displayed on first row of the LCD. On pressing the same button to select the message (say, Contact me) it appears on second row of the LCD. Now, press ES1 (send button) to send the message. Confirmation of message dispatch will then appear on the LCD followed by time and date.

Data packet format sent by the Module1 is: *Ascii27+receiver_id_no+sender_id_no+sender_user_name+space+time+ascii22+the_message_string+ascii26+null*

Receiving a message

Module2, which has the user ID_NO selected by Module1 for sending, receives the data, processes it, and displays on the LCD screen continu-